

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) now deliver striking performance in image, video, and other generative tasks, yet they still produce hallucinated samples (*hallucinations*) that violate real-world knowledge, such as an implausible duplicate cup floating beside another cup. This failure mode is no longer anecdotal: the absence of practical, systematic quantification methods directly limits progress on factual generative modeling and obscures how next-generation models should be designed under explicit factual constraints. We address this gap by isolating and formalizing a concrete, measurable failure mode that we call **counting hallucination**: the generation of an incorrect number of object instances or structured parts, such as a hand with six fingers, despite the absence of such patterns in the training data. We introduce **CountHalluSet**, a dataset suite with explicit counting criteria, consisting of ToyShape, SimObject, and RealHand. On this suite, we establish a standardized evaluation protocol for counting hallucinations and use it to systematically determine how solver type, ODE solver order, sampling steps, and initial noise shape hallucination rates in DPMs. Our analyses further show that common perceptual metrics, including Fréchet Inception Distance (FID), do not consistently capture counting hallucinations, establishing the need for dedicated factuality-oriented evaluation.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become a leading generative modeling paradigm, producing high-quality samples for text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Yet this progress has not solved factual generation: diffusion models still fail to reliably preserve factual properties encoded in the training data and can generate outputs that contradict real-world consistency. A particularly revealing failure is incorrect counting of spatially intricate components, including extra human fingers (Narasimhaswamy et al., 2024) and additional animal limbs (Borji, 2023).

Recent work has begun to expose and organize the failure modes of text-guided diffusion models. For example, Borji (2023) cataloged qualitative shortcomings in text-to-image models, Liu et al. (2024) introduced adversarial search to uncover undesirable generation behaviors, and Wu et al. (2023) developed an iterative procedure for discovering and mitigating spurious correlations. These studies make clear that failure analysis is now essential for understanding diffusion models, but they do not yet provide a targeted quantitative framework for factual hallucinations.

Hallucination in diffusion models has therefore become a central failure mode to characterize (Aithal et al., 2024). Unlike text-prompt alignment failures, which can be evaluated through input-output consistency measures such as CLIP score (Hessel et al., 2021) or VQA score (Huang et al., 2025; Zarei et al., 2024), and unlike obvious visual defects such as severe distortion, blur, or artifacts, hallucinated samples (*hallucinations*) often look plausible while violating factual consistency with the training data. They include incorrect object counts, geometries that appear locally plausible but are physically impossible, and other violations of physical regularities. These subtle distinctions between valid samples and hallucinations expose a decisive weakness in current AI systems, which often cannot detect such failures reliably. For example, even state-of-the-art large vision-language models such as GPT-4o can struggle to count fingers in a clear hand image. A standardized and well-defined

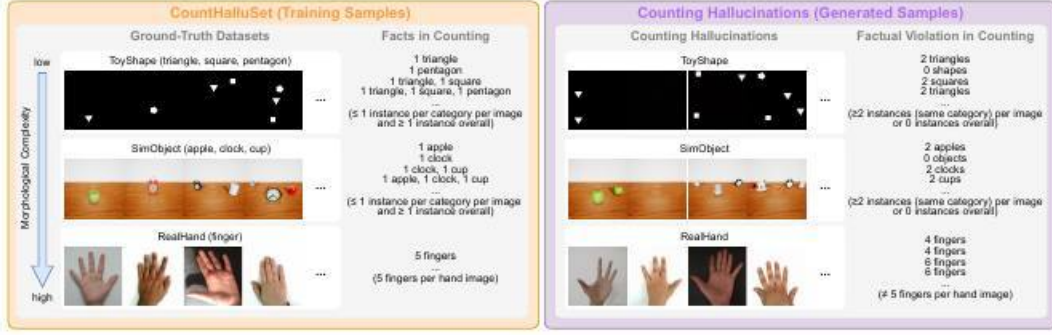


Figure 1: Representative comparison between **CountHalluSet** training data (left) and model-generated **counting hallucinations** (right). Each dataset enforces explicit counting criteria: ToyShape and SimObject contain at most one instance per category and at least one instance overall, whereas RealHand contains exactly five fingers. Counting hallucinations are generated images that violate these rules, such as two apples in one image or an empty scene with a normal background, even though such patterns never appear in SimObject.

protocol for quantifying hallucinated samples is therefore necessary: it supplies the measurement foundation required to design generative models with strict factual consistency, a capability that is especially important if such models are to serve as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among hallucinated attributes—including counts, colors, spatial relationships, geometry, and physical realism—counting errors provide the cleanest entry point for systematic study because they are visually interpretable and directly measurable. We therefore define a specific and operational failure mode, **counting hallucination**. *In this setting, the number of objects in the training data is explicitly predetermined. A generated image is hallucinated exactly when its object counts deviate from these predefined values.* Thus, a hand image with six fingers is a counting hallucination because this pattern is absent from RealHand, as shown in the last row of Fig. 1. To make this phenomenon measurable across increasing visual complexity, we construct CountHalluSet with three datasets: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 presents representative training examples and the corresponding hallucinations that our protocol targets.

Accurate object identification is the central technical requirement for quantifying counting hallucinations, so our protocol uses counting models tailored to each dataset. With these datasets and models in place, we directly measure how denoising conditions—sampling steps, initial noise, ODE solver order, and solver type—change counting hallucination rates in diffusion models. We also test whether standard perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021), actually track counting factuality.

Our main findings and contributions are:

- We show that standard numerical improvements in diffusion sampling, including increasing sampling steps from 25 to 100, do not reliably reduce counting hallucinations for human hands, even though they help on simpler synthetic datasets.
- We establish that counting hallucination rate is only weakly and inconsistently associated with perceptual metrics, while non-counting failure rate is strongly associated with those metrics.
- Motivated by these findings, we introduce joint-diffusion models, a training paradigm that substantially reduces both counting hallucinations and non-counting failures on RealHand.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

The growing literature on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023) has sharpened attention on analogous failures in diffusion models.

For unconditional diffusion models, Aithal et al. (2024) proposed the qualitative hypothesis that hallucinated samples may arise from interpolation between distinct data modes. This explanation is valuable, but the field still lacks the quantitative machinery needed to measure hallucination rates and compare failure severity across sampling conditions.

Our work supplies this missing measurement layer by turning hallucination identification into an explicit quantification problem for diffusion-generated outputs. This framing enables direct analysis of how denoising choices, including sampling steps and ODE solver order, affect hallucination severity rather than merely sample appearance.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are widely used for evaluating text-guided image generation, but language-model-based evaluation is poorly suited as the sole arbiter of hallucinated visual content. A statistical lower bound on the tendency of pretrained language models to hallucinate certain fact types limits the detection accuracy achievable by language models alone (Kalai & Vempala, 2024). The persistent gap between human annotators and LLMs in classifying hallucinated text provides further evidence that hallucination evaluation requires task-specific measurement rather than generic language-based scoring (Lin et al., 2022; Li et al., 2023).

Image generation is also commonly evaluated with quantitative perceptual metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). Many of these metrics summarize feature embeddings from a pretrained Inception-v3 model (Szegedy et al., 2016), and they are effective for measuring broad perceptual fidelity. What they do not guarantee is sensitivity to factual violations. We address this unresolved issue directly by testing these metrics against counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a, *diffusion process*) that progressively destroys data structure by adding Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on the original data $\mathbf{x}_0$ is:

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a pre-defined, timestep-dependent noise schedule, typically linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021). The factor $\bar{\alpha}_t$ controls the signal-to-noise ratio: $\bar{\alpha}_0 = 1$ gives clean data, while $\bar{\alpha}_T \approx 0$ gives nearly pure noise. Equivalently, Eq. (1) can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the marginal distribution $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Widely used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021) typically set T to 1000, making this Gaussian endpoint the starting point for generation.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) is the generative mechanism that reconstructs data from diffused noise $\mathbf{x}_T$. DPMs learn this process by predicting the noise $\boldsymbol{\epsilon}$ in $\mathbf{x}_t$ with a neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, the *noise prediction model*. The parameter θ is trained by minimizing the standard objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\pi(t) \|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, ancestral sampling generates samples from $q_0(\mathbf{x}_0)$ by reversing the diffusion process with the same number of steps:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where the only unknown quantity is $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$, the *score function* (Song et al., 2022).

In practice, the noise prediction model approximates the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$. Together with Eq. (3), this yields the equivalent update:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Ancestral sampling in Eq. (4) is stochastic (Kloeden et al., 1992), but it has a deterministic counterpart formulated as an ordinary differential equation (ODE), also known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory satisfying this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and initialized from ideal diffused data $\mathbf{x}(T)$ drawn from $q_T(\mathbf{x}(T))$. Set the discrete time steps to $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. The exact one-step solution follows from the variation-of-constants formula:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear component $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$. The step size $\Delta t > 0$ allows ODE-based samplers to accelerate generation by taking larger steps than ancestral sampling, with $T/\Delta t$ total sampling steps. Because $\mathcal{G}(t_2, t_1)$ is available in closed form, accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical design—higher solver order and additional sampling steps—on approximating the nonlinear integral involving h . Appendix A.1 to A.3 provides the theoretical details that motivate our experimental focus on these solver choices.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section provides the full measurement framework for counting hallucinations. We first formalize counting criteria and define hallucinations as violations of those criteria, then introduce CountHalluSet. For each dataset, we state the counting rules, identify the corresponding hallucination cases, and specify the standardized quantification protocol.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We begin with a general definition that turns counting hallucination into a precise dataset-relative event: a generated sample is hallucinated when it violates the counting constraints that hold in the reference training set.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote the reference training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the count of category c , denoted $N_c(\mathbf{x})$, must belong to a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), so $N_c(\mathbf{x}) \in \mathcal{S}_c$. We also require every valid reference sample to contain at least one object, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we classify it as a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ exactly when the following two conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{x}$ must have sufficient visual quality for the background and any present objects to be clearly discernible. This condition cleanly separates counting hallucinations from coarse failures such as severe degradation. We define the binary *counting-ready indicator* $\mathbb{I}_{\text{CRI}}(\hat{x}) \in \{0, 1\}$, where 1 means the image is suitable for counting and 0 means it is not. In practice, this indicator can be provided by human annotation, by a classifier calibrated to human annotations, or by another automated criterion.
2. **Violation of counting facts.** A counting-fact violation occurs when some category c has $N_c(\hat{x}) \notin S_c$, or when the image contains no objects, $\sum_{c \in C} N_c(\hat{x}) = 0$, despite the dataset requirement that every sample contain at least one object. We operationalize this test with a counting model that predicts object counts in generated images.

Thus, a counting hallucination (CH) is present when the image is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{x}) = 1$) and the predicted counts violate the criteria ($\exists c : N_c(\hat{x}) \notin S_c$ or $\sum_{c \in C} N_c(\hat{x}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{x}) = \mathbb{I}_{\text{CRI}}(\hat{x}) \wedge \left(\exists c : N_c(\hat{x}) \notin S_c \vee \sum_{c \in C} N_c(\hat{x}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

CountHalluSet consists of three datasets designed to expose counting hallucinations at increasing levels of visual complexity: ToyShape, SimObject, and RealHand. Examples from the datasets and their corresponding hallucinations appear in Fig. A1 and A2. Appendix A.4 provides additional statistics.

4.2.1 TOYSHAPE

Dataset description. ToyShape contains 30,000 images spanning three geometric categories: *triangle*, *square*, and *pentagon*. Every shape is white, has area 120 pixels, and appears without overlap, giving a deliberately controlled testbed for counting. The counting criteria are explicit: each image contains at most one instance of each category and at least one shape overall, as shown in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(x) \in S_c = \{0, 1\}$, subject to $\sum_{c \in C} N_c(x) \geq 1$.

Counting hallucinations. Under these criteria, any ToyShape generation with two or more instances of the same category is a counting hallucination, such as the image with two pentagons in the top-right region of Fig. 1. Empty generations containing only the black background are also counting hallucinations because they violate the at-least-one-object rule. Appendix A.5 provides examples.

Counting model used in ToyShape. We identify ToyShape counting errors with a dedicated counting model and do not require a counting-ready indicator, because severe degradation is rare in this controlled dataset. Concretely, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape images where each category appears 0–3 times per sample (i.e., $S_c = \{0, 1, 2, 3\}$), covering both counting-valid samples and hallucination patterns needed for reliable quantification. The resulting model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For each diffusion model trained on ToyShape, we generate 30,000 samples, matching the training-set size. The counting model predicts the object counts in every generated image, and any sample that violates the ToyShape criteria is labeled as a counting hallucination, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. SimObject contains 30,000 rendered images of everyday objects and raises the visual complexity beyond ToyShape while preserving exact counting rules. It includes three categories—*mug*, *apple*, and *clock*—with 10 intra-class variants per category. Objects are randomly placed on a wooden tabletop under fixed lighting with minimal mutual occlusion, as shown in the middle-left of Fig. 1. We synthesize the dataset in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering of geometry, material attributes, lighting, and reflections. The counting criteria match the ToyShape structure (see Sec. 4.2.1). Formally, $C = \{\text{mug}, \text{apple}, \text{clock}\}$. For each class $c \in C$, we set $N_c(x) \in S_c = \{0, 1\}$, subject to $\sum_{c \in C} N_c(x) \geq 1$.

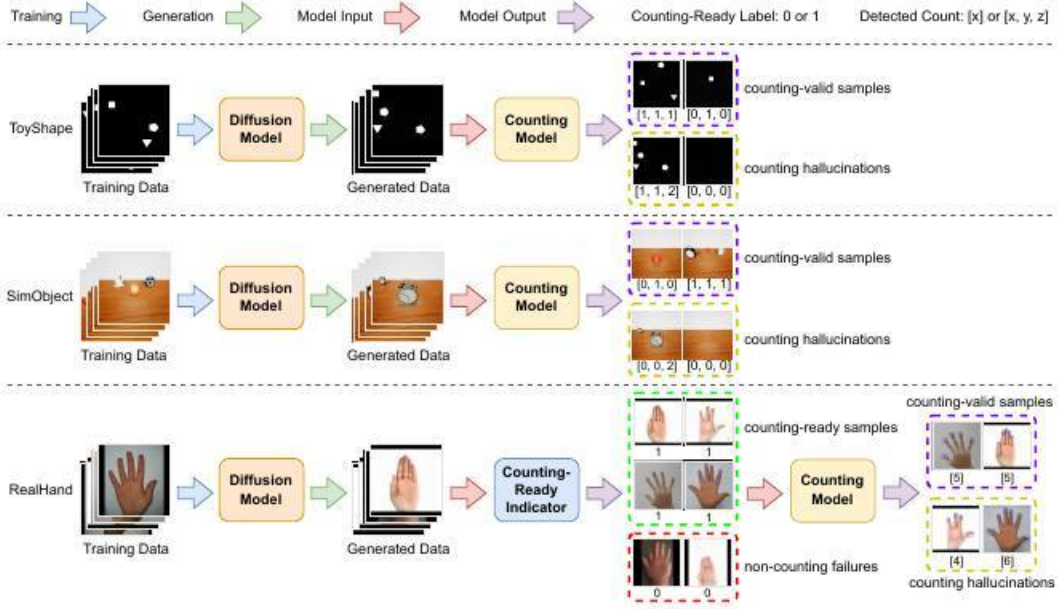


Figure 2: CountHalluSet evaluation pipeline. ToyShape and SimObject generations are evaluated directly by counting models. RealHand first uses a counting-ready indicator, implemented as a binary classifier aligned with human judgments, to separate counting hallucinations from non-counting failures such as severely deformed fingers. Count labels are ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

Quantification of counting hallucinations. Because SimObject uses the same counting structure as ToyShape, its hallucination definition, counting-model construction, and quantification protocol follow the same procedure, as summarized in Fig. 2.

4.2.3 REALHAND

Dataset description. RealHand brings the evaluation to real-world morphology by collecting 5,050 human hand images from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate the raw images and remove samples with blur or artifacts. Each retained image shows a human hand with five fingers, covering dorsal (back) and palmar (front) views under varied lighting and backgrounds such as white surfaces, blackboards, and walls, as shown in the lower-left of Fig. 1. The counting criterion is strict: every image contains exactly five fingers, i.e., $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. Realistic hand generation also produces failures that are not countable hallucinations, including blur, severe finger deformation, and visual artifacts. We therefore use a counting-ready indicator, implemented as a binary classifier, to separate images that can be meaningfully counted from these non-counting failures. Appendix A.7 shows examples of both counting hallucinations and non-counting failures.

Counting-ready indicator and counting model used in RealHand. To operationalize this separation, we fine-tune MaxViT (Tu et al., 2022) on 2.5k annotated generated images as a counting-ready indicator aligned with human judgments. For fingertip counting, we fine-tune YOLO-12 (Tian et al., 2025) on 2k generated images with fingertip annotations. Human evaluation on 500 randomly sampled images by three annotators confirms that the indicator and counting model achieve over 96% and 99% accuracy, respectively. Appendix A.6 provides YOLO detection examples.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where all generated images can be counted directly because severe failures are rare, RealHand requires explicit filtering

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

before counting. We first apply the counting-ready indicator to retain hand images with sufficient visual quality and then use the counting model to predict finger counts, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. We train the diffusion backbones under a controlled setup that isolates the effect of denoising choices on counting hallucinations. For ToyShape, we train a DDPM (Ho et al., 2020) from scratch; for SimObject and RealHand, we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on CelebA-HQ (Karras et al., 2018). We set T to 1,000 with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and to 256×256 for LDMs. All models use Adam (Kingma, 2015) with learning rate 0.0001 and are trained for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively, with effective batch size 256.

Inference of Diffusion Models. For ancestral sampling (DDPM), we use 1,000 sampling steps. For ODE-based sampling, we evaluate both a first-order solver, DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)), and a second-order solver, DPM-Solver-2 (Lu et al., 2022). Each ODE solver is tested at the common denoising budgets of 25, 50, and 100 steps. For RealHand, the YOLO counting model uses a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless otherwise specified, each evaluation generates as many samples as the corresponding training set, and all reported results are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across solver types, ODE solver orders, sampling steps, and initial-noise settings for ToyShape, SimObject, and RealHand. The results reveal five clear patterns.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand.** As Table 1 shows, increasing ODE sampling steps from 25 to 50 to 100 generally lowers counting hallucination rates on ToyShape and SimObject, but increases them on RealHand except for the 100-step DPM-Solver-2 setting. This contrast demonstrates that finer solver granularity benefits simpler, highly structured synthetic distributions, while on RealHand additional denoising can amplify local inconsistencies in a more complex real-world distribution.

(2) **Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps.** As shown in Table 1, comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling budgets, especially 25 and 50 steps, exposes a robust trade-off: the higher-order solver substantially reduces non-counting failures while increasing counting hallucination rate. Increasing the number of sampling steps produces the same qualitative tension for both solvers. Thus, spending more computation on nominally stronger sampling improves perceptual quality, reflected by lower NCFR, but does not enforce factual correctness of finger counts.

(3) **DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across all comparisons, ancestral sampling (DDPM) consistently achieves the lowest CHR, NCFR, and TFR. This establishes DDPM as the strongest configuration in our evaluation and a practical reference point for lower failure rates under the tested generative settings.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, “diffused” noise (ground-truth initial noise) usually yields lower CHR, NCFR, and TFR than “normal” noise (standard Gaussian noise). This pattern indicates that better alignment with the training-time initial-noise distribution can reduce both counting hallucinations and broader non-counting failures.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that counting hallucinations rise with the morphological complexity of the counted objects. Fine-grained details

Table 1: Failure-rate comparison across solver configurations and initial-noise settings on ToyShape, SimObject, and RealHand. CHR is the number of counting hallucinations divided by the total number of generated samples. For RealHand, NCFR measures non-counting failures detected by the counting-ready indicator, and TFR is CHR plus NCFR. *Diffused* and *Normal* denote ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

Table 2: Correlations between failure rates and perceptual metrics, including FID and MUSIQ. PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to generate 5050 samples, matching RealHand. PTI-Hand FID is computed between RealHand images and model-generated samples. Here, p denotes the p-value for the statistical significance of r or ρ . $\dagger$ indicates that DDPM results are integrated with results from ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01	-0.97	<0.01	-0.99	<0.01	-0.99	<0.01
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—	-0.98	<0.01	—	—	-0.98	<0.01
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29	-0.88	<0.01	-0.46	0.29	-0.88	<0.01
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—	-0.91	<0.01	—	—	-0.91	<0.01
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—	0.97	—	0.99	—	0.99	—
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—	0.97	—	—	—	—	—

and complex spatial relationships make accurate count preservation harder for diffusion models, producing more frequent additions and omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because counting hallucination is a factuality failure, a central question is unavoidable: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our evidence answers **not necessarily**.

We test this relationship by correlating CHR and NCFR with the same perceptual metrics. The analysis covers FID (Heusel et al., 2017), a distribution-level metric, and MUSIQ (Ke et al., 2021), an instance-level quality metric. We use Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The results separate factual counting failures from perceptual failures.

✎ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** The first and third rows of Table 2 show that CHR–FID correlations are not stable across datasets: Pearson correlation is positive for SimObject, negative for RealHand, and positive again for PTI-Hand, while Spearman correlation shifts from positive to negative to negative. Distance correlation also drops from 0.82 on SimObject to 0.74 on RealHand and 0.47 on PTI-Hand, showing that more complex generative content weakens the CHR–FID association. When DDPM results are integrated with ODE-solver results, these correlations weaken further.

✎ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast, NCFR correlates strongly and robustly with perceptual metrics across settings. For NCFR vs. FID, all reported coefficients exceed 0.88 and have very small p-values, confirming that these metrics capture perceptual degradation far more reliably than counting factuality.

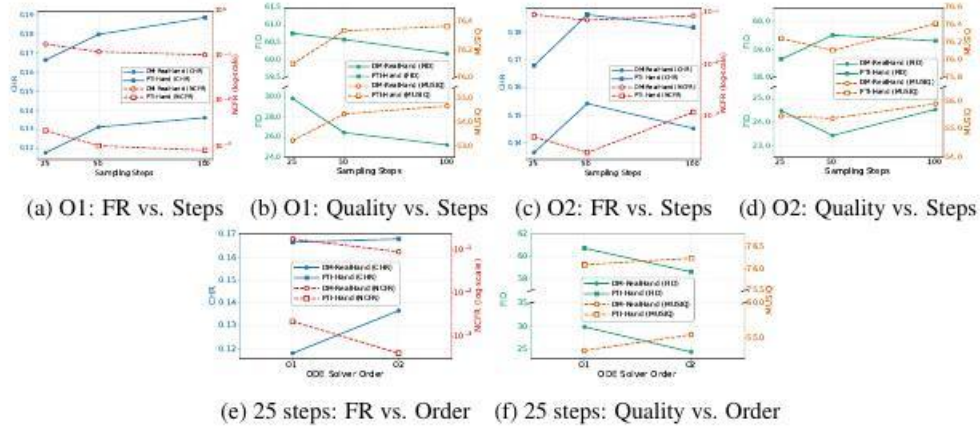


Figure 3: Trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCFR, while quality metrics include FID and MUSIQ. Across sampling-step and solver-order changes, the two models show strongly aligned trends.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We next test whether the RealHand findings transfer to a large-scale pretrained text-to-image model. Specifically, we evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model from Sec. 4.2.3; Appendix A.8 gives the details. As Figure 3 shows, solver type and sampling steps affect counting hallucinations in the pretrained model in a way that closely matches our controlled RealHand experiments. This agreement validates the main trends in Sec. 5.2 and demonstrates that the evaluation protocol applies beyond models trained only on CountHalluSet. The correlation results in Table 2 reinforce this conclusion: CHR remains weakly related to perceptual metrics, whereas NCFR remains strongly related to them. Together, these results establish the robustness and broader relevance of our findings for pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

Observation 5 in Sec. 5.2 identifies a concrete path forward: diffusion models preserve counting facts more reliably when the countable visual structure has low morphological complexity, as with masks. This motivates a direct intervention: denoise RGB representations jointly with explicit structural constraints, such as object masks that encode topology and counts, in a shared latent space. We call this training paradigm **joint-diffusion models** (JDM), because the model learns general visual representations and structural constraints together during denoising.

Concretely, we concatenate the latent representation of each hand mask channel-wise with the latent representation of its corresponding RGB image, as shown in Fig. 4. We then fine-tune the CelebA-HQ-pretrained UNet (Karras et al., 2018), using the RealHand configuration from Sec. 5.1, on RealHand images and their SAM-2-extracted masks (Ravi et al., 2024). To support both modalities, we expand the denoising UNet input and output channels to match the combined latent dimensionality of image and mask latents (i.e., $3+3=6$ channels).

Table 3 shows that JDM consistently lowers both CHR and NCFR relative to the vanilla LDM across sampling conditions, demonstrating that explicit structural denoising improves generation of count-sensitive hand structure. JDM does increase FID, which is consistent with the added denoising burden of jointly modeling RGB images and masks in a small-scale training regime. This trade-off is informative rather than contradictory: structural factuality improves even when the perceptual metric worsens. We also observe encouraging reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under controlled fine-tuning settings. Appendix A.9 provides additional details, results, and discussion.

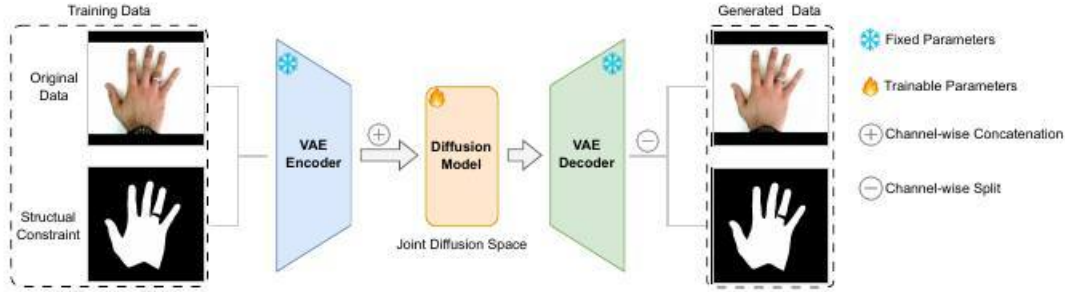


Figure 4: Proposed joint-diffusion model (JDM). JDM concatenates latent representations of hand RGB images and their corresponding segmentation masks channel-wise. Joint denoising supplies explicit structural information, encouraging anatomically plausible hand poses and correct finger topology, which reduces both CHR and NCFR.

Table 3: Failure-rate comparison between the vanilla LDM and the proposed JDM on RealHand across ODE solvers and sampling steps. Lower values are better. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs to preserve direct comparability with baselines.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

6 CONCLUSION

This work establishes a concrete measurement foundation for hallucinations in diffusion models. By formalizing counting hallucination, introducing **CountHalluSet**, and providing a reproducible evaluation protocol, we turn a visually salient but previously under-quantified failure mode into a measurable target. Our controlled experiments show that standard numerical improvements in sampling often fail to reduce hand counting hallucinations, even when they improve perceptual quality. Our correlation analyses further demonstrate that widely used perceptual metrics such as FID and MUSIQ do not consistently reflect counting hallucination severity. Finally, experiments on large-scale pretrained text-to-image models confirm that the main trends generalize beyond the controlled training setting, while JDM demonstrates a practical route for reducing count-based failures through explicit structural denoising.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the learned reverse process $p_\theta(\mathbf{x}_{T:0})$, we define the KL divergence as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Using the Markov property, both reverse processes decompose cleanly as:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule gives the following decomposition of the KL divergence:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

The Data Processing Inequality (Cover & Thomas, 2012) then guarantees that marginalizing, or discarding random variables, cannot increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error follows directly by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Equality (=) holds only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for every t and $\mathbf{x}_t$; in that case, the diffusion model recovers the underlying data distribution $q(\mathbf{x}_0)$ exactly. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and generated data matches the DDPM variational bound (Ho et al., 2020) after excluding data entropy $H(\mathbf{x}_0)$. Thus, optimizing the DDPM variational loss directly tightens an explicit bound on the discrepancy between true data and model-generated data.

Diffused-prior gap. The first error source is the mismatch between the ground-truth diffused distribution $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the sampling prior $p_\theta(\mathbf{x}_T)$. This mismatch is unavoidable in practice because the number of diffusion steps T is finite. A general distributional metric quantifies the discrepancy as:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes Kullback–Leibler (KL) divergence. Other appropriate metrics, including FID or Wasserstein distance, can also evaluate this distributional discrepancy. Because the diffused-prior gap is present at initialization, it propagates through denoising and yields persistent output error.

Noise model prediction error. The second error source is the noise prediction model itself. Although $\epsilon_\theta(\mathbf{x}_t, t)$ can approximate the true noise ϵ in principle under universal approximation results (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022), finite models and finite training leave residual prediction error. Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this error. From Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Building on Eq. (7), we analyze a one-step numerical solver as the basic case for accumulated ODE sampling error. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies the ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from the ground-truth initial condition $\mathbf{x}^*(T)$ drawn from $q_T(\mathbf{x}_T)$. Let $\hat{\mathbf{x}}_k$ denote the numerical trajectory produced by an ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the estimated model function. For a specific numerical trajectory from $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$, the objective is to express its total accumulated trajectory error. Define $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ as the trajectory error at step k . We analyze one transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. The ideal one-step evolution, moving backward from τ_k to τ_{k+1} , is given by the variation-of-constants formula:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear component $\frac{dz}{d\tau} = f(\tau)z$. The integration limits are intentionally backward in time.

Numerical path evolution. A one-step numerical solver approximates this evolution, and its update deviates from the exact trajectory governed by $\hat{h}$ starting at $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} of the ODE using $\hat{h}$, initialized at $\hat{\mathbf{x}}_k$ at time τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method contributes a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Therefore, the numerical step is:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12) gives the error recurrence:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. Unrolling this recurrence from $k = 0$ to $N - 1$ gives the accumulated error. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Repeated application yields:

$$\begin{aligned} e_N &= \mathcal{G}(\tau_N, \tau_0) e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \\ &\quad \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \end{aligned} \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ yields:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

The propagated initial error term captures the deterministic contribution of a specific initial error instance to the final path error; statistically, this term originates from the diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), DDPM (Ho et al., 2020) uses the following forward process:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

Its reverse process is:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

DDIM (Song et al., 2021) shares the same forward process as Eq. (A.16), but its deterministic reverse process (with $\eta = 0$) is:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\theta}(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_{\theta}(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_{\theta}(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Thus, although DDPM and DDIM share the same forward diffusion process, DDIM is not merely DDPM with the stochastic reverse term removed. The two methods use different coefficient formulations for the predicted noise term, which is why their reverse dynamics differ.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from all three datasets. Table A1 confirms that each dataset is balanced across its categories, and Table A2 shows comparable distributions over object counts. These statistics verify that CountHalluSet controls category and count bias rather than relying on incidental imbalances. The datasets appear in the same order in both tables, and the category order in Table A1 is:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Category-level object counts for each CountHalluSet dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Image counts grouped by object count for each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

Fig. A2 provides representative counting hallucinations from all three datasets. For ToyShape and SimObject, hallucinations either contain multiple instances of at least one category or contain no instances despite a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples have invalid finger counts, such as 4 or 6 fingers, as shown in Fig. A2(c). The last two images in the first row of Fig. A2(c) may appear to show five fingers, but one finger is partially incomplete and visibly artifacted, leaving only four valid fingers. They are therefore correctly classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on generated RealHand images. The detector reliably identifies fingertips across diverse poses and orientations, including dorsal and palmar views. However, pure fingertip counting cannot distinguish some anatomically implausible but five-finger configurations, such as duplicated thumbs in Fig. A3(a) or five middle fingers without a thumb in Fig. A3(a), from anatomically valid five-finger hands. These counting-valid but anatomically abnormal samples therefore remain indistinguishable under the count criterion alone.

Figure A3(b) shows counting-hallucinated samples where YOLO predicts invalid fingertip counts, including 3, 4, 6, and 7. Figure A3(c) highlights rare edge cases (<1%) involving unclear boundaries, cracked fingers, or floating artifacts. Adjusting YOLO inference hyperparameters, such as IoU and confidence thresholds, can mitigate some of these cases, but it can also degrade detection on the much larger set of counting-valid samples. Because the same detection setup is used throughout RealHand evaluation, these rare cases do not alter the paper’s conclusions.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows the three RealHand generation classes used by our protocol: counting-valid, counting-hallucinated, and non-counting failure. As illustrated in Fig. 2, RealHand evaluation first applies the counting-ready indicator to separate countable samples from images unsuitable for counting. Unsuitable samples typically contain severe artifacts, including distorted, cracked, partially visible, or floating fingers. The remaining counting-ready samples are classified as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

For PTI-Hand, we use 10 hand-generation prompts with Stable-Diffusion-3.5-Medium. The prompts are:

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.

6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Counting hallucination is defined relative to patterns absent from training data, but this absence cannot be fully verified for large-scale pretrained diffusion models with inaccessible training distributions. We therefore adopt a conservative PTI-Hand criterion: any generated hand image with six or more fingers is counted as a hallucination. This criterion is justified because real images may show fewer than five visible fingers due to occlusion, articulation, or viewpoint, but hands with six or more fingers are highly unlikely in ordinary real-image training data. Finger counts ≥ 6 therefore provide a reliable and empirically verifiable hallucination signal. Generated examples appear in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM experiments in Table 3, we fine-tune the denoising UNet for 120k steps, compared with 80k steps for the vanilla LDM. This larger budget is necessary because expanding the latent space from three to six channels creates a substantial mismatch with the pretrained LDM latent distribution. Additional optimization is therefore required to close this distributional gap and realize JDM’s structural modeling advantage.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Failure-rate and image-quality comparison between fine-tuned SD-1.5 and SD-1.5 fine-tuned with the joint-diffusion training paradigm on RealHand. The first row reports the baseline obtained by fine-tuning Stable-Diffusion-1.5 on RealHand. The second row freezes all intermediate layers of the baseline model and fine-tunes only the input and output layers with extended channel dimensions. Lower values indicate better performance.

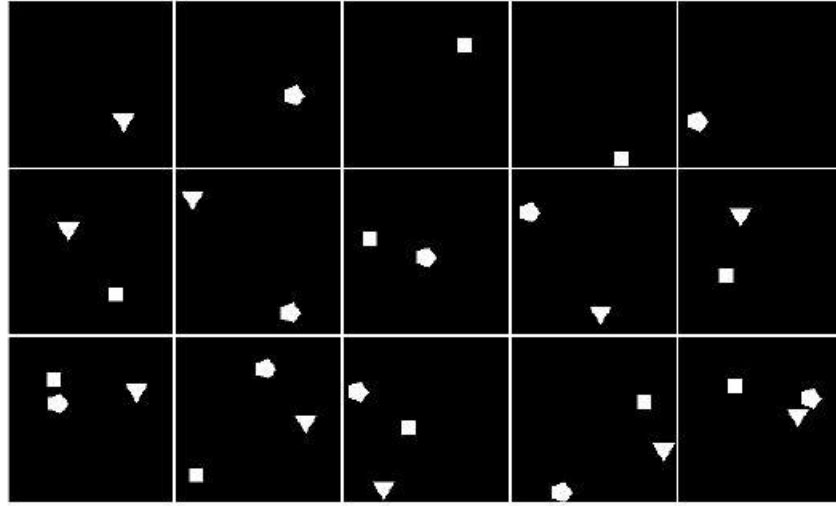
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

JDM should be understood as a **novel training paradigm** for improving factual generation in diffusion models, not as a claim of universally superior inference. It extends only the UNet input and output channels while leaving intermediate layer sizes unchanged, so it **introduces negligible inference-time overhead compared to vanilla LDM**. At the same time, adding mask channels to a pretrained text-to-image model **disrupts the manifold geometry of the pretrained latent space and changes the original denoising distribution**, making direct fine-tuning nontrivial.

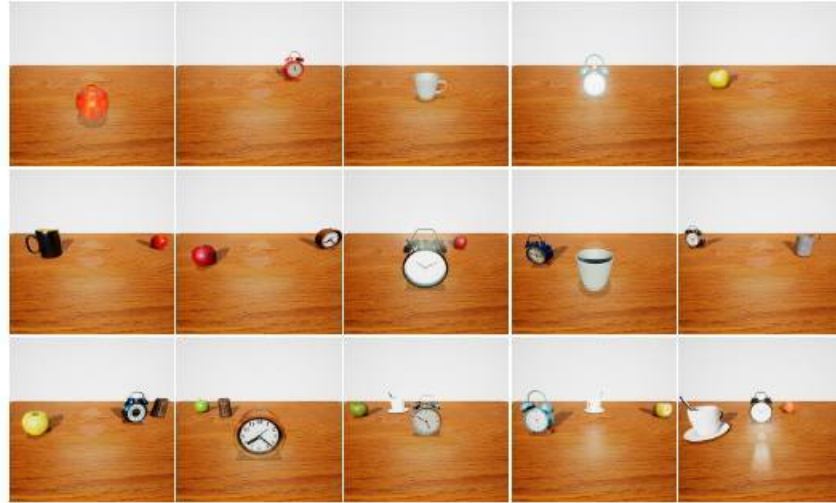
Despite this difficulty, JDM still reduces counting hallucinations in large pretrained diffusion models under carefully controlled fine-tuning. The central challenge is aligning JDM’s extended latent space with the rigid latent geometry of a large pretrained model using only the limited RealHand dataset. We therefore use a two-stage strategy: first fine-tune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and fine-tune only the input and output layers with extended channels. As Table A3 shows, even this constrained setup lowers counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence supports JDM as a strong direction for enforcing factual consistency in image generation. A decisive next step is **exploring its behavior and performance under true large-**

972 **scale training**, where the structural benefits observed here can be tested without the constraints of
973 small-scale fine-tuning.
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(a) Representative ToyShape examples

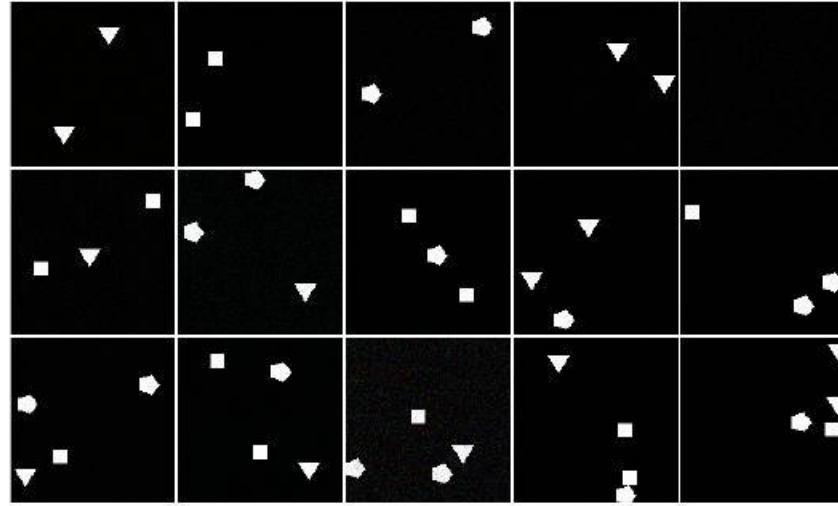


(b) Representative SimObject examples



(c) Representative RealHand examples

Figure A1: Representative datasets in CountHalluSet: ToyShape, SimObject, and RealHand.



(a) ToyShape counting hallucinations

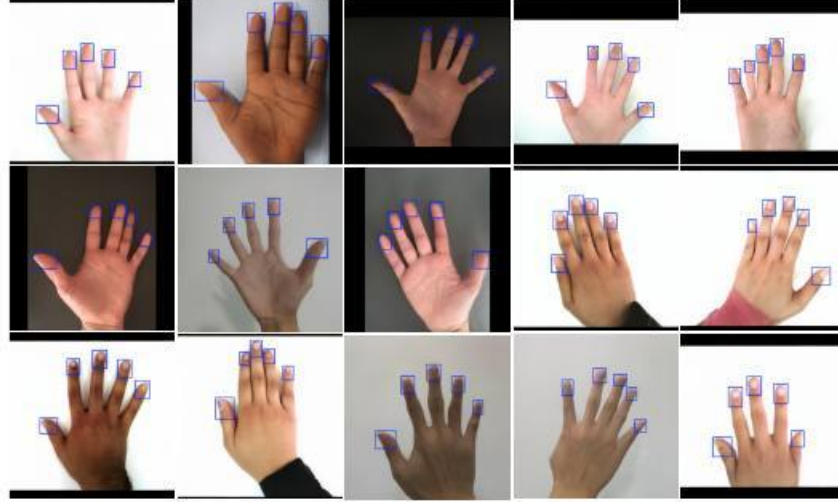


(b) SimObject counting hallucinations

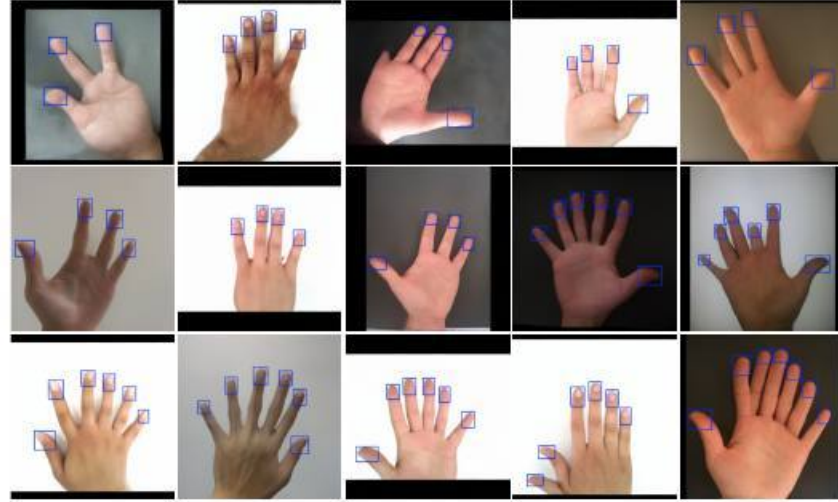


(c) RealHand counting hallucinations

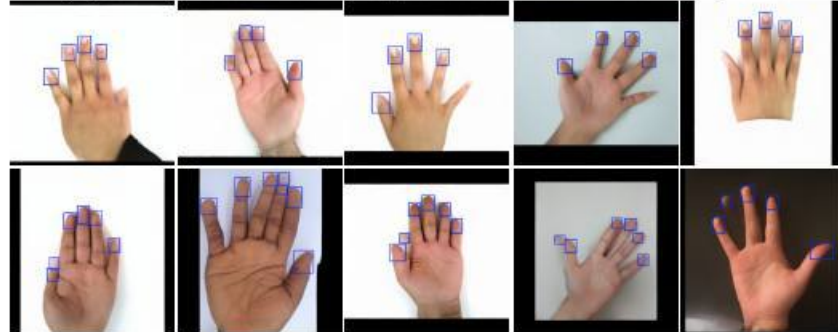
Figure A2: Representative counting hallucinations from ToyShape, SimObject, and RealHand, illustrating the dataset-specific violations targeted by CountHalluSet.



(a) Counting-valid YOLO predictions with five detected fingers



(b) Counting-hallucinated YOLO predictions with an invalid fingertip count

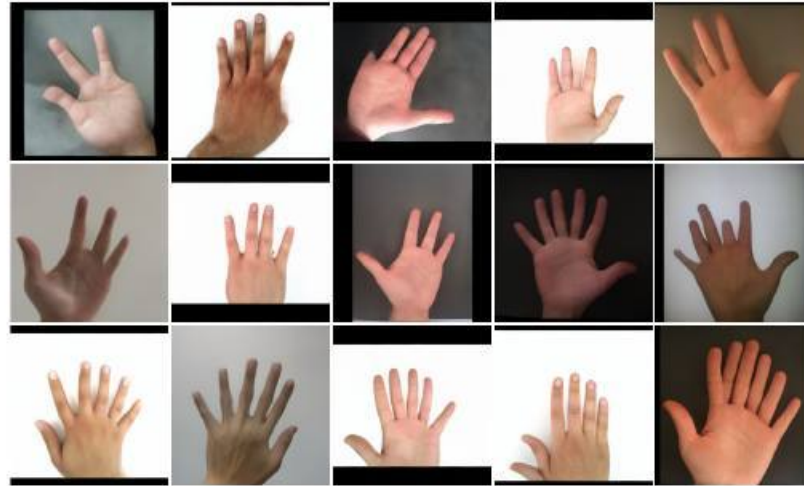


(c) Rare YOLO failure cases

Figure A3: YOLO predictions on generated RealHand images. (a) Counting-valid samples where YOLO correctly predicts five fingertip boxes, including cases with two thumbs. (b) Counting-hallucinated samples where YOLO predicts an invalid fingertip count, such as 3, 4, 6, or 7 fingertips. (c) Rare edge cases that the counting-ready indicator does not filter and that remain difficult for YOLO, including blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with an invalid number of valid fingers



(c) Non-counting failures unsuitable for counting

Figure A4: Three classes of generated RealHand images. (a) Counting-valid samples contain five clearly discernible fingers. (b) Counting-hallucinated samples contain an incorrect number of valid fingers, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe artifacts, including distorted, cracked, partially visible, or floating fingers.

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(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with more than five fingers



(c) Non-counting failures unsuitable for counting

Figure A5: Three classes of hand images generated by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid samples contain five clearly discernible fingers. (b) Counting-hallucinated samples contain more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe artifacts, including distorted, cracked, partially visible, or floating fingers.